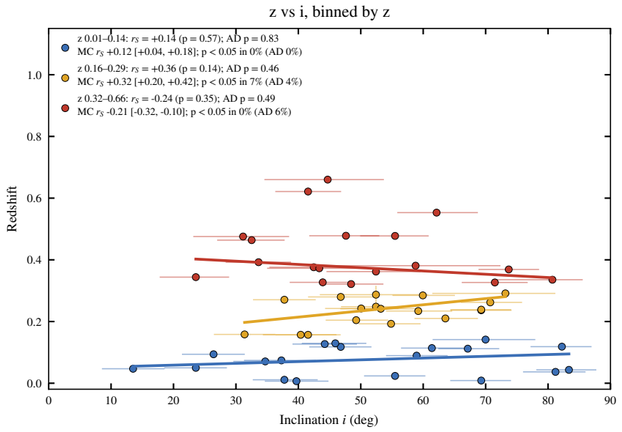
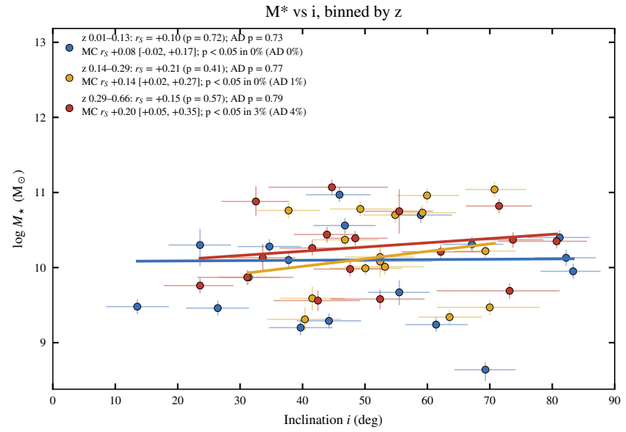
z, M*, SFR and DM_host against inclination within tertiles of each quantity

z vs i, binned by z



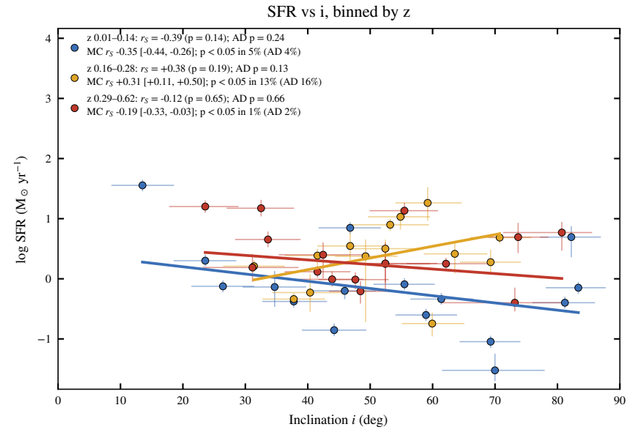
all 55: r_S = -0.02 (p = 0.86); AD p = 0.53; partial τ: binned on the y quantity
pooled within tertiles: r = +0.09 (p = 0.54); tertiles agree: p = 0.25
through the errors (MC median [16, 84]%): all r_S -0.04 [-0.09, +0.01], p < 0.05 in 0% (AD 0%)

M* vs i, binned by z



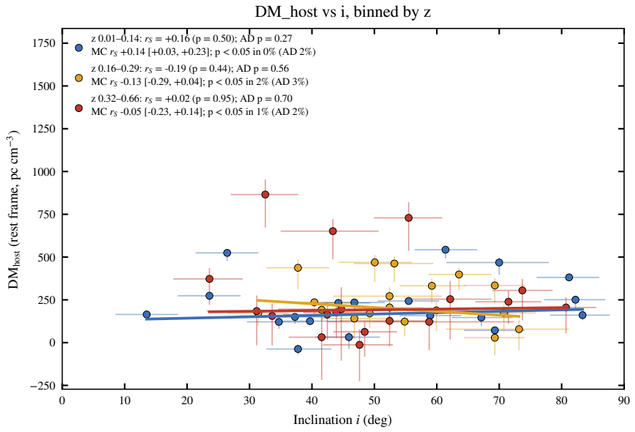
all 51: r_S = +0.14 (p = 0.31); AD p = 0.99; partial τ | z = +0.11 (p = 0.21)
pooled within tertiles: r = +0.15 (p = 0.33); tertiles agree: p = 0.95
through the errors (MC median [16, 84]%): all r_S +0.13 [+0.07, +0.19], p < 0.05 in 0% (AD 1%)
MC partial τ +0.09 [+0.06, +0.14]

SFR vs i, binned by z



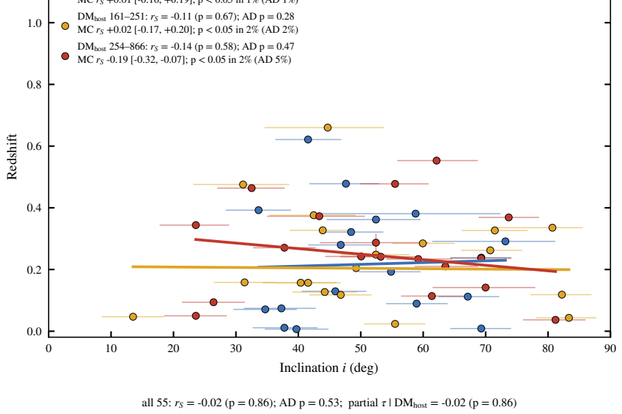
all 46: r_S = -0.12 (p = 0.42); AD p = 0.46; partial τ | z = -0.06 (p = 0.56)
pooled within tertiles: r = -0.07 (p = 0.68); tertiles agree: p = 0.16
through the errors (MC median [16, 84]%): all r_S -0.14 [-0.20, -0.08], p < 0.05 in 1% (AD 0%)
MC partial τ -0.09 [-0.13, -0.05]

DM_host vs i, binned by z



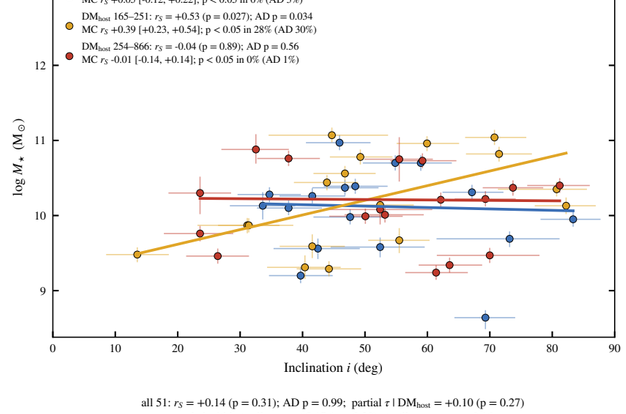
all 55: r_S = +0.06 (p = 0.64); AD p = 0.19; partial τ | z = +0.05 (p = 0.64)
pooled within tertiles: r = -0.00 (p = 0.99); tertiles agree: p = 0.62
through the errors (MC median [16, 84]%): all r_S +0.02 [-0.06, +0.11], p < 0.05 in 0% (AD 3%)
MC partial τ +0.01 [-0.04, +0.07]

z vs i, binned by M*



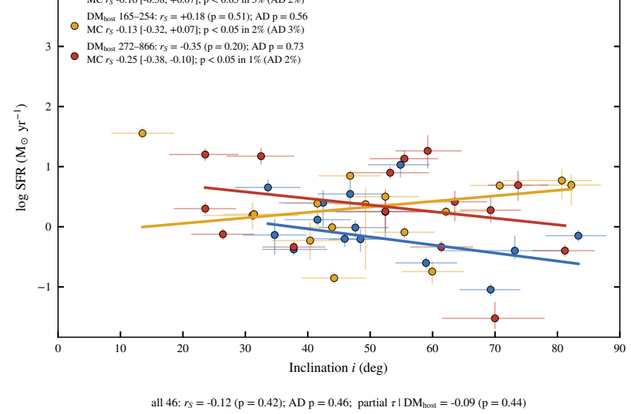
all 51: r_S = -0.04 (p = 0.76); AD p = 0.57; partial τ | log $M_\star$ = -0.05 (p = 0.61)
pooled within tertiles: r = -0.12 (p = 0.45); tertiles agree: p = 0.85
through the errors (MC median [16, 84]%): all r_S -0.05 [-0.10, -0.01], p < 0.05 in 0% (AD 0%)
MC partial τ -0.05 [-0.09, -0.02]

M* vs i, binned by M*



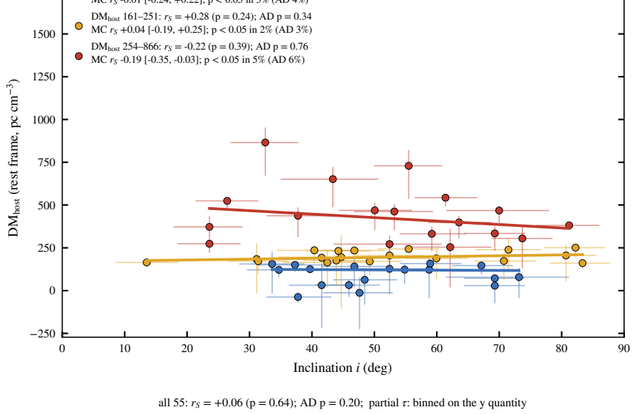
all 52: r_S = +0.11 (p = 0.43); AD p = 1.00; partial τ: binned on the y quantity
pooled within tertiles: r = +0.03 (p = 0.86); tertiles agree: p = 0.64
through the errors (MC median [16, 84]%): all r_S +0.10 [+0.04, +0.16], p < 0.05 in 0% (AD 0%)

SFR vs i, binned by M*



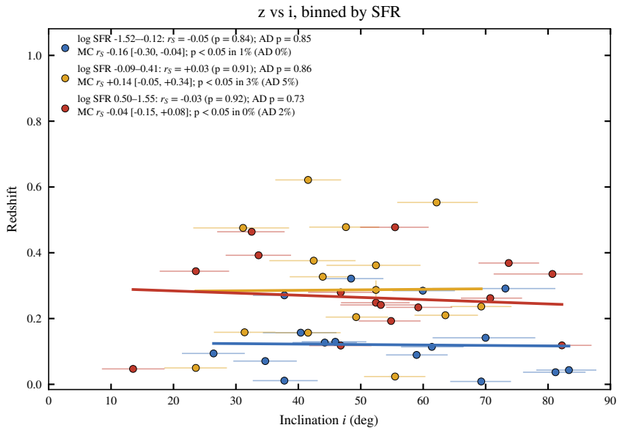
all 46: r_S = -0.12 (p = 0.42); AD p = 0.46; partial τ | log $M_\star$ = -0.08 (p = 0.36)
pooled within tertiles: r = -0.18 (p = 0.29); tertiles agree: p = 0.054
through the errors (MC median [16, 84]%): all r_S -0.14 [-0.20, -0.08], p < 0.05 in 1% (AD 0%)
MC partial τ -0.11 [-0.15, -0.07]

DM_host vs i, binned by M*



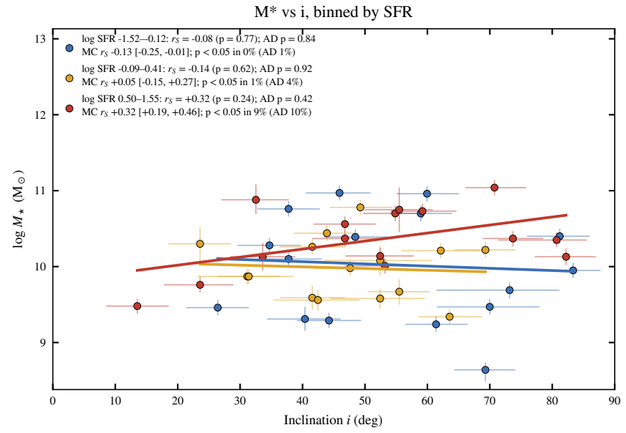
all 51: r_S = +0.12 (p = 0.39); AD p = 0.043; partial τ | log $M_\star$ = +0.09 (p = 0.38)
pooled within tertiles: r = +0.09 (p = 0.55); tertiles agree: p = 0.68
through the errors (MC median [16, 84]%): all r_S +0.06 [-0.02, +0.15], p < 0.05 in 1% (AD 6%)
MC partial τ +0.05 [-0.01, +0.10]

z vs i, binned by SFR



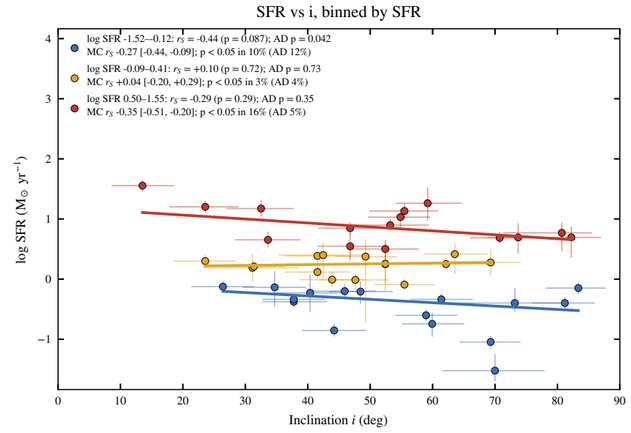
all 46: r_S = -0.07 (p = 0.65); AD p = 0.75; partial τ | log SFR = -0.03 (p = 0.75)
pooled within tertiles: r = -0.02 (p = 0.92); tertiles agree: p = 0.98
through the errors (MC median [16, 84]%): all r_S -0.07 [-0.12, -0.03], p < 0.05 in 0% (AD 0%)
MC partial τ -0.02 [-0.05, +0.01]

M* vs i, binned by SFR



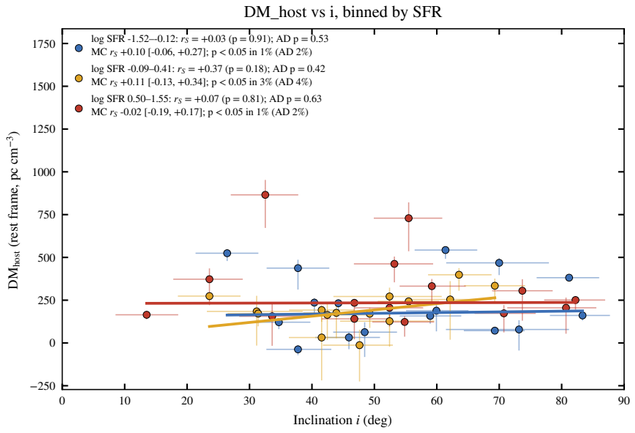
all 46: r_S = +0.10 (p = 0.52); AD p = 0.99; partial τ | log SFR = +0.09 (p = 0.38)
pooled within tertiles: r = +0.03 (p = 0.84); tertiles agree: p = 0.47
through the errors (MC median [16, 84]%): all r_S +0.09 [+0.03, +0.15], p < 0.05 in 0% (AD 0%)
MC partial τ +0.08 [+0.04, +0.12]

SFR vs i, binned by SFR



all 46: r_S = -0.12 (p = 0.42); AD p = 0.46; partial τ: binned on the y quantity
pooled within tertiles: r = -0.23 (p = 0.17); tertiles agree: p = 0.36
through the errors (MC median [16, 84]%): all r_S -0.14 [-0.20, -0.08], p < 0.05 in 1% (AD 0%)

DM_host vs i, binned by SFR



all 46: r_S = +0.12 (p = 0.42); AD p = 0.079; partial τ | log SFR = +0.10 (p = 0.28)
pooled within tertiles: r = +0.16 (p = 0.35); tertiles agree: p = 0.64
through the errors (MC median [16, 84]%): all r_S +0.07 [-0.02, +0.16], p < 0.05 in 1% (AD 6%)
MC partial τ +0.06 [-0.00, +0.12]

z vs i, binned by DM_host

all 55: r_S = -0.02 (p = 0.86); AD p = 0.53; partial τ | DM$_{host}$ = -0.02 (p = 0.86)
pooled within tertiles: r = -0.06 (p = 0.67); tertiles agree: p = 0.86
through the errors (MC median [16, 84]%): all r_S -0.04 [-0.09, +0.01], p < 0.05 in 0% (AD 0%)
MC partial τ -0.02 [-0.06, +0.01]

M* vs i, binned by DM_host

all 51: r_S = +0.14 (p = 0.31); AD p = 0.99; partial τ | DM$_{host}$ = +0.10 (p = 0.27)
pooled within tertiles: r = +0.17 (p = 0.29); tertiles agree: p = 0.16
through the errors (MC median [16, 84]%): all r_S +0.13 [+0.07, +0.19], p < 0.05 in 0% (AD 1%)
MC partial τ +0.09 [+0.05, +0.13]

SFR vs i, binned by DM_host

all 46: r_S = -0.12 (p = 0.42); AD p = 0.46; partial τ | DM$_{host}$ = -0.09 (p = 0.44)
pooled within tertiles: r = -0.19 (p = 0.25); tertiles agree: p = 0.26
through the errors (MC median [16, 84]%): all r_S -0.14 [-0.20, -0.08], p < 0.05 in 1% (AD 0%)
MC partial τ -0.10 [-0.15, -0.06]

DM_host vs i, binned by DM_host

all 55: r_S = +0.06 (p = 0.64); AD p = 0.20; partial τ: binned on the y quantity
pooled within tertiles: r = +0.00 (p = 0.98); tertiles agree: p = 0.36
through the errors (MC median [16, 84]%): all r_S +0.02 [-0.06, +0.11], p < 0.05 in 0% (AD 3%)

How to read each legend entry: tertile range of the binning quantity; Spearman rank correlation r_S of the plotted pair in that tertile, with its two-sided p-value (null hypothesis: $r_S = 0$).
AD: two-sample Anderson–Darling p-value that the y quantity has the same distribution for hosts below and above the median of the x quantity (5,000 permutations); it also detects non-monotonic dependence.
MC r_S: the same Spearman coefficient recomputed in 2,000 draws in which every value is resampled within its error bars and the tertiles are re-formed; median [16th, 84th percentile] of r_S (not a p-value), and the fraction of draws with Spearman p < 0.05 (in brackets: with AD p < 0.05).